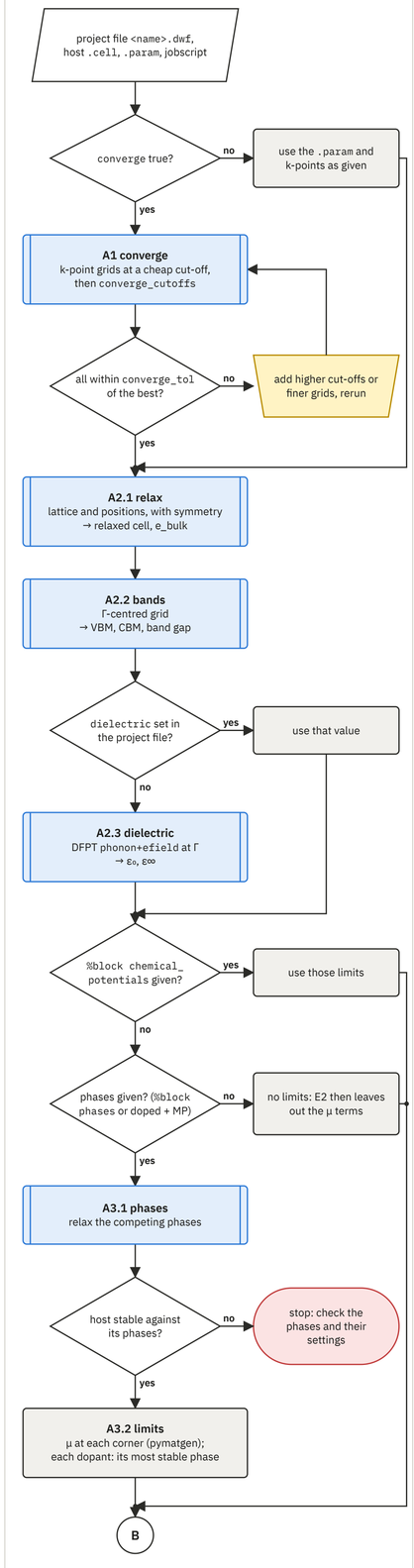
Defect workflow, as dwf.py runs it

One column per phase, top to bottom; each diamond is a decision and every branch out of it is labelled. One command runs every phase in order: `python3 dwf.py <name>.dwf`; each phase keeps its results.json (read by the later phases) and summary.txt in its folder.

start / end · step done by dwf.py · CASTEP jobs (supercell_convergence.py) · MACE (charge-blind; runs where dwf.py runs) · ShakeNBreak · your action · decision · input · B on to the next phase

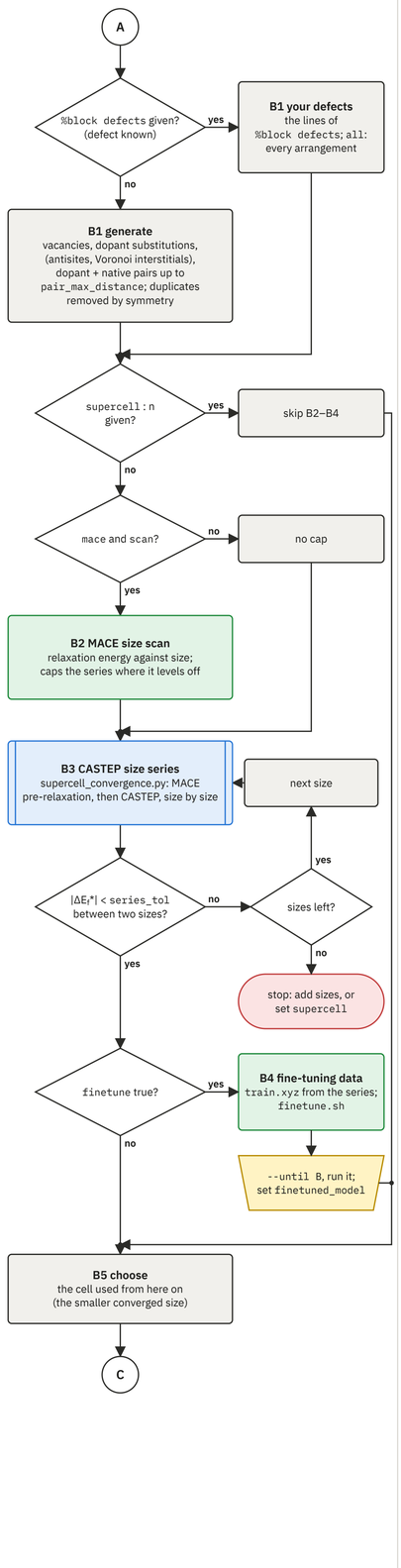
A Host reference

A_host/ · once per host



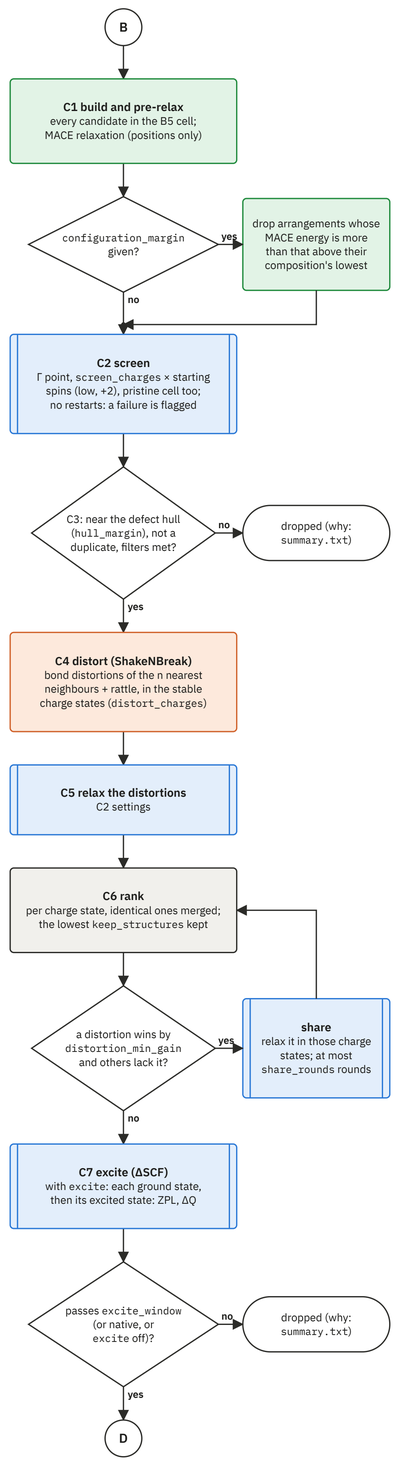
B Candidates and cell size

B_candidates/ · once per host



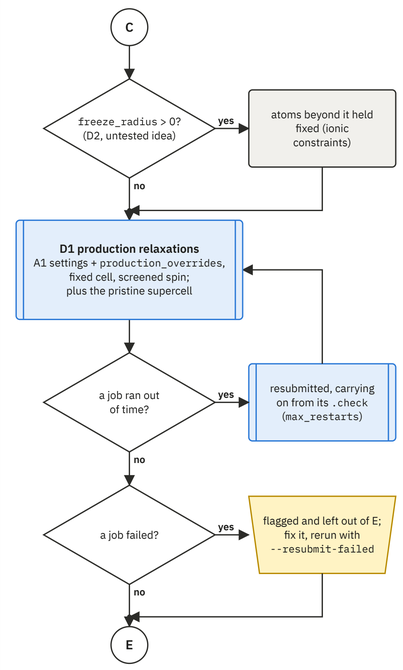
C Screening and ground states

C_screen/ · every candidate, then survivors



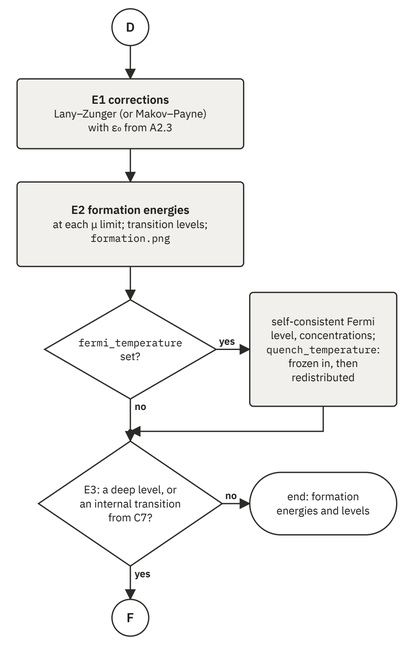
D Production relaxations

D_production/ · survivors × charge states



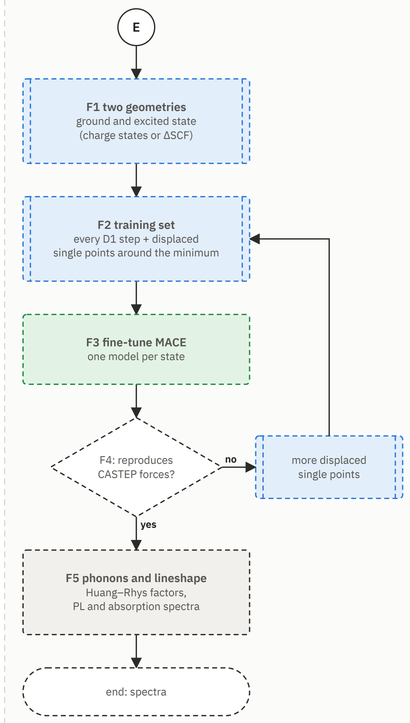
E Thermodynamics

E_thermo/ · per defect



F Optical lineshape

not scripted yet · selected defects



Every CASTEP step: the job runner

`supercell_convergence.py` batch mode, for each step:

- one dry run per distinct input, then a plan per job: nodes, memory, partition, walltime
- at most `max_jobs` queued at once
- out of time: carried on from the `.check` (`max_restarts`; 0 for screening)
- `tolerate_failures`: a failed job is flagged and the others carry on (screening, production)
- a job whose inputs changed since it ran stops the run, so results from two settings never mix
- `touch <workdir>/STOP` stops it; jobs keep running

Checkpoints and reruns

`<workdir>/state.json` records each finished step with a fingerprint of its settings, files and earlier results.

- stopped (STOP, Ctrl-C, `--no-wait`, a crash)? Rerun the same command: finished steps are skipped, submitted and finished jobs reused
- changed a setting? The steps that use it, and those after them, run again; unchanged jobs are reused
- `--redo STEP`: that step from scratch (old files kept)
- `--status`: what has finished; `--until B`: stop after B
- `--prepare-only`: write the inputs, submit nothing

Phase F is not scripted yet

E3 lists the defects worth a lineshape (`for_F` in `E_thermo/results.json`); each D1 relaxation's `.castep` (every step) is training data for F2. Fine-tuning and phonons: MACE and lineshape_tools (Turiansky et al.).

Charge-aware MACE: not in the workflow yet

Every MACE step in this chart (green) is charge-blind: the model sees no charge and no spin. It only prepares structures (B2, B3, C1) and never ranks charge states, so CASTEP does all the searching in C and D, and F needs one fine-tuned model per charge state.

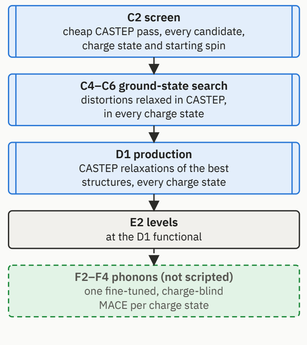
A charge-aware MACE (Wang, Mosquera-Lois and Walsh, arXiv:2603.05238, March 2026):

- a learnable total-charge embedding in MACE (v0.3.15): one model covers all five charge states of a Se vacancy in Sb_2Se_3
- multi-fidelity: PBE relaxations, plus HSE06 single points on about 10 % of them
- transition levels within 0.02 eV of HSE06; exploring the energy surface about three orders of magnitude cheaper
- models with explicit charges (MACE-POLAR, MACE-LES) did worse there, and the molecular MACE-POLAR-1 missed the DFT ground states

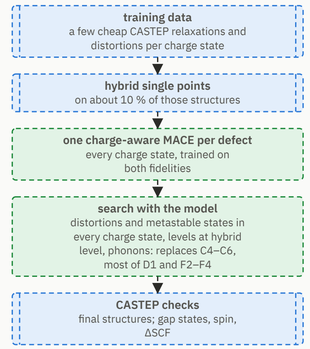
What would stay. CASTEP makes the training data and has the last word: gap states, spin and excited states still need it. A model trained on one defect is reliable only near the structures it has seen, so a minimum it finds far from them is a candidate to check in CASTEP, not a result.

Status. One preprint and one test material; its data and trained weights "will be made available upon publication". It used VASP (PAW); with CASTEP, the PBE and hybrid runs would need the same pseudopotentials and settings.

Now: dwf.py



With a charge-aware, multi-fidelity MACE



What the filters in C do

- C1 `configuration_margin`: drop arrangements more than this above the lowest MACE energy of their composition (charge-blind)
- C3 hull: $E_f(E_F)$ of each arrangement and charge state; kept if within `hull_margin` of its group's lowest at some E_F
- C3 duplicate: dropped if it relaxed into the structure of a lower one
- C3 properties (optional), in a charge state stable at some E_F: `require_gap_states`, `require_spin`, `require_bound`
- C6 `keep_structures`: the n lowest per charge state go on

C7: excited states (`excite : true`)

- ground state: a single point at the screening settings (`metals_method : dm`); its `.check` is kept
- excitation: the smallest occupied → empty step ≥ `excite_min` between gap levels, in one spin channel
- excited state: CASTEP's simple ΔSCF from that `.check`; relaxed (`excite_relax`): the ZPL and ΔQ, else the vertical energy
- verdict: pass (in `excite_window`), outside, none, or unknown (its runs failed: kept, with a warning)
- native defects always go on, and skip C3's property filters

Test C7 on one defect first

CASTEP's ΔSCF must work on your build: try Sb_{Ca}–V_O in q = −1 (ADAQ: ZPL about 0.53 eV). If its excited state won't relax, set `excite_relax : false`.